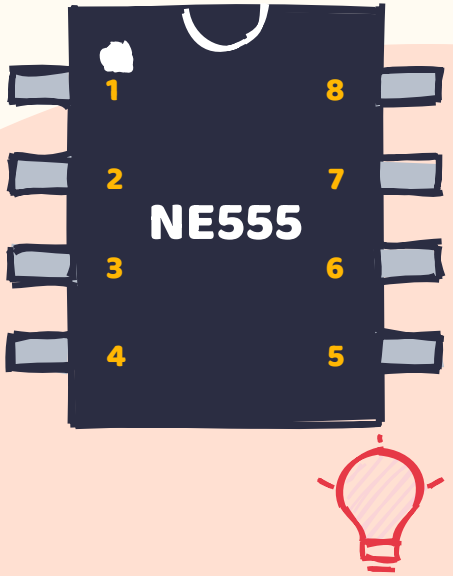
Young Innovators · Week 1

555 Timer Alarm Siren Circuit

Build a flashing, beeping alarm with no code at all

Ages 12-14 **No microcontroller** **Falstad simulator**



Did you know?
The 555 was designed in 1971. Billions have been made and it is still used today!

Look and listen

Blink, beep, blink, beep...



Car alarms

Flash and beep in a steady rhythm



Ambulance lights

Blink on and off, again and again



Kitchen timers

Beep when time is up

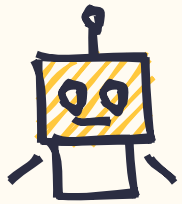
Many simple devices have no computer and no code. So what makes them keep perfect time?

Learning aims

By the end of this session, I can...

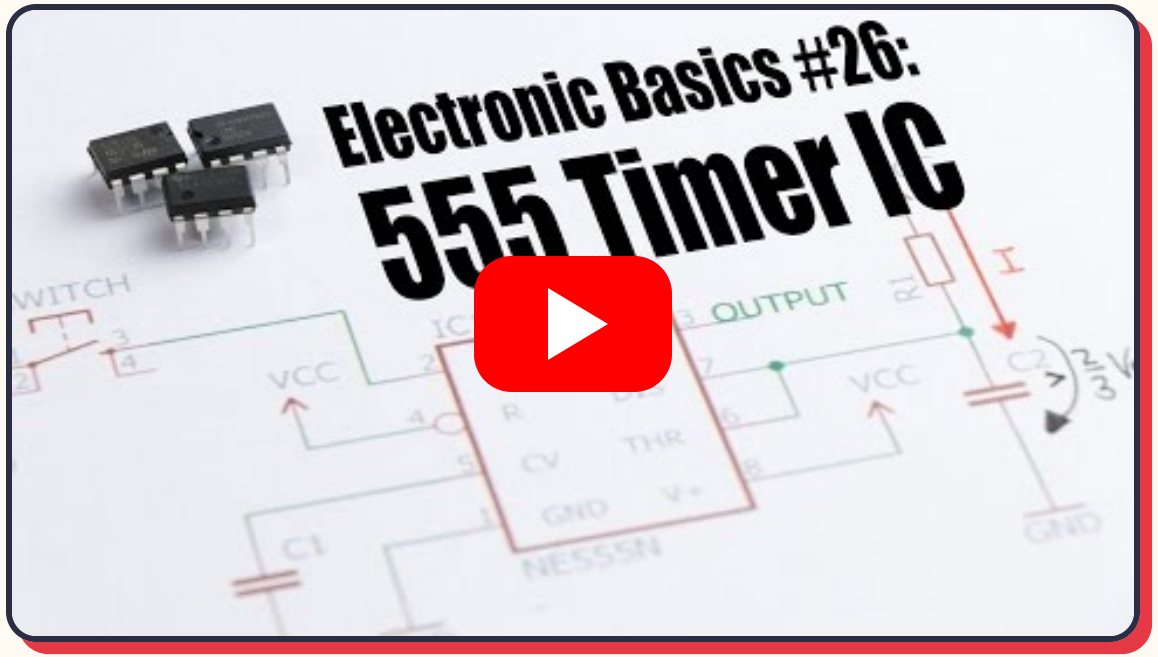
1. Explain what an **integrated circuit** is and identify the 555's 8 pins
2. Explain how a **capacitor** charging and discharging creates timing
3. Build a 555 **astable** alarm circuit on a breadboard
4. Measure voltages safely with a **multimeter**
5. Calculate and change the **frequency** using R and C

Keywords
IC · capacitor · charge · astable · frequency · square wave · multimeter



Watch and think

Video time!



Electronic Basics #26: 555 Timer IC

GreatScott! · about 8 minutes · youtu.be/fLaexx-NMj8

Watch for...

1. What does a 555 timer make in **astable** mode?
2. Which parts set how **fast** it blinks?
3. Why is the 555 still used after 50 years?

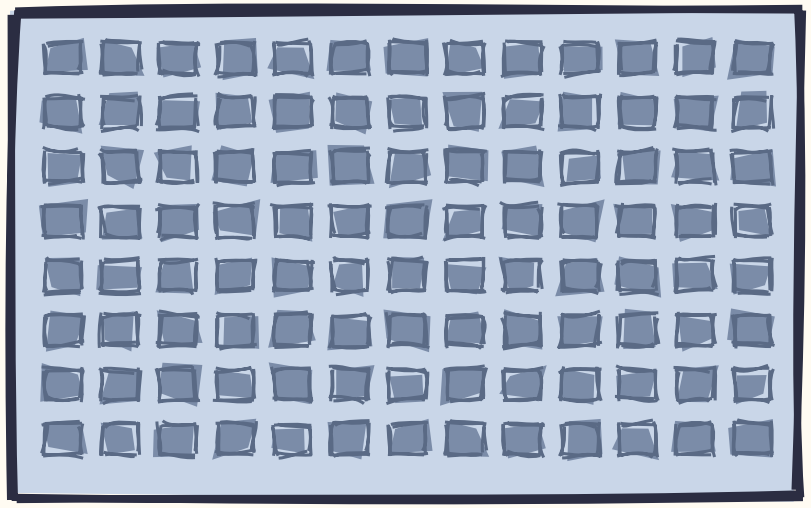
Scan to watch

After the video: Predict: if we use a bigger capacitor, will our siren beep faster or slower? Why?

LEARN

What is a chip?

Integrated circuits (ICs)



tiny transistors and resistors on silicon

A whole circuit in one tiny package

An **integrated circuit** puts many transistors, resistors and connections onto one piece of **silicon**.

The 555 has about **25 transistors** inside. Your phone chip has **billions**.

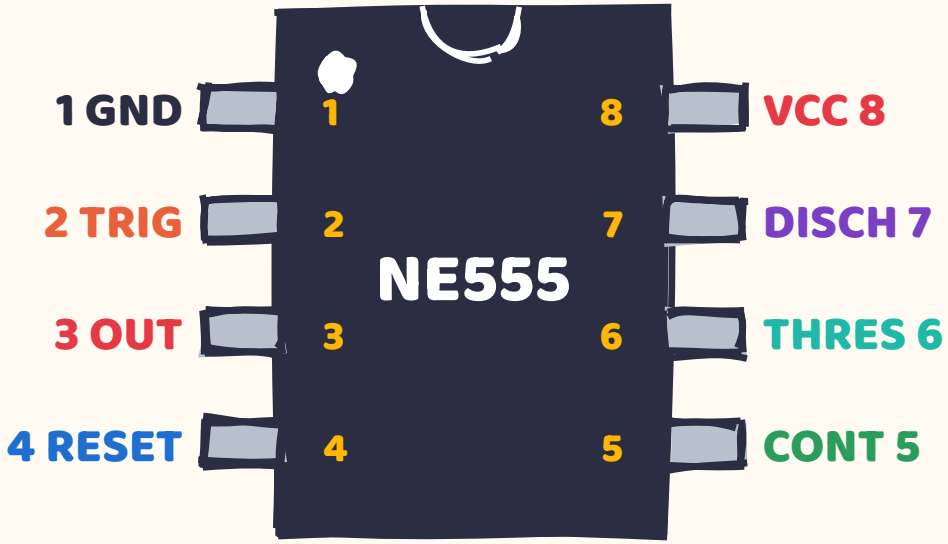
In real life

ICs are inside everything electronic: watches, cars, washing machines, traffic lights and toys.

LEARN

Meet the star

The 555 timer: 8 pins, 8 jobs



Find pin 1

Look for the **notch** or **dot**. Pin 1 is next to it. Count **anti-clockwise** round the chip.

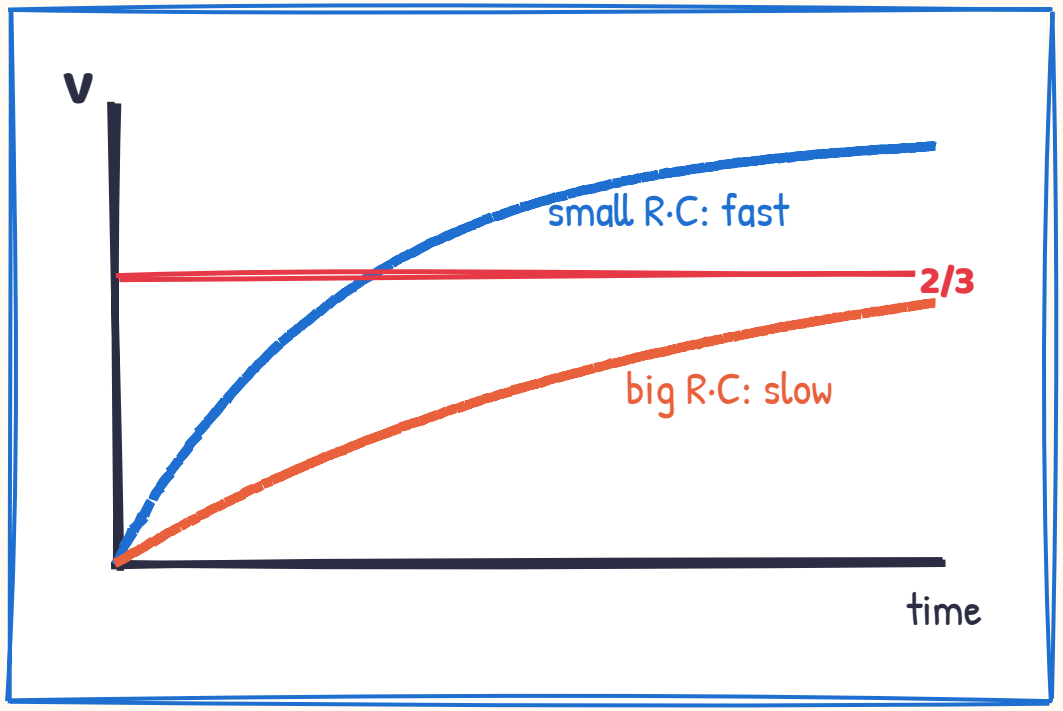
Pins we use today

8 power + · **1** ground
3 output (LED + buzzer)
2, 6, 7 timing (R and C)
4 reset (to +) · **5** 10nF to ground

LEARN

The timing secret

Capacitors: tiny charge buckets



Fill, then empty

A capacitor stores electric charge, like a bucket filling with water through a tap.

Bigger resistor = slower tap.

Bigger capacitor = bigger bucket.

Both make the timing **slower**.

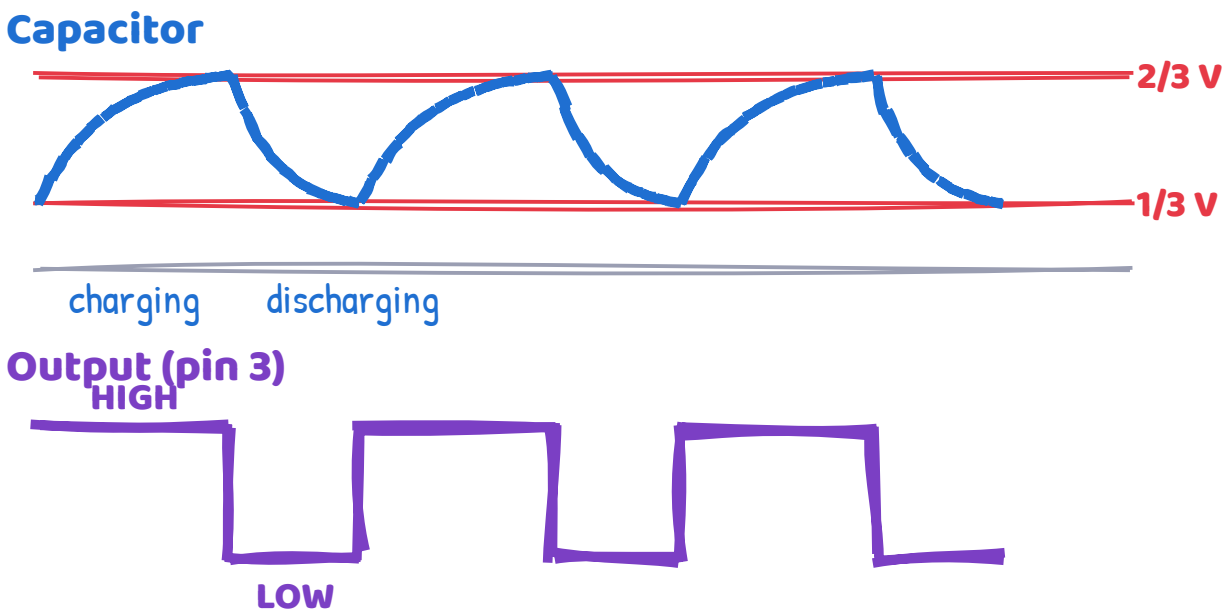
In real life

A camera flash charges a capacitor, then empties it in a split second for a bright flash.

LEARN

How it works

Astable mode: a loop that never stops

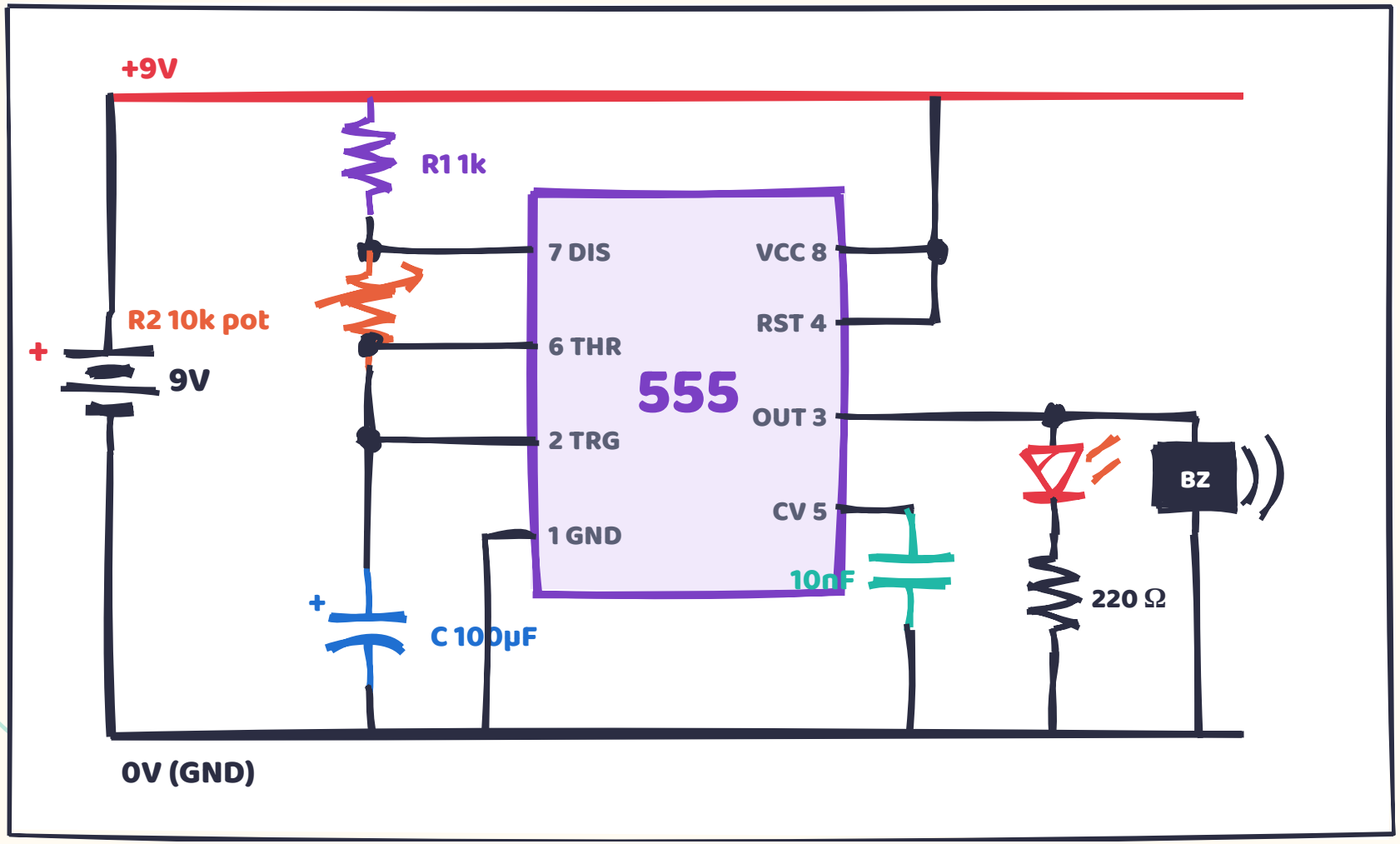


1. The capacitor charges through R1 and R2. Output is **HIGH**: LED on, buzzer beeps.
2. At **2/3** of the supply, the 555 flips. It discharges through R2. Output **LOW**.
3. At **1/3**, it flips back. Repeat forever!

The output is a **square wave**: ON, OFF, ON, OFF... "astable" means it never settles.

The plan

Our alarm circuit (schematic)



Read the colours

Purple = timing resistor R1. Orange = speed knob R2. Blue = timing capacitor.

Teacher tip

Check the buzzer label: use one rated for 9V (many are 3–12V), or it may overheat.

LEARN

Measure like an engineer

Multimeter checks

Measure	Expected	Your reading
Battery	about 9 V	_____
Pin 8 to pin 1	about 9 V	_____
Pin 3 (slow flash)	jumps high / low	_____
Capacitor (slow flash)	rises 3 V to 6 V	_____

How to measure

Dial to **20 V DC**. Black probe on **-** rail. Touch the red probe on each pin.
Use the **100 µF** version so the numbers change slowly enough to read.

Think
Why does the capacitor stay between 3 V and 6 V on a 9 V supply? (Hint: 1/3 and 2/3!)

Why it matters

The 555 in the real world



Car indicators

Older cars used timer circuits like this to blink



Toys and doorbells

Tones and melodies



Motor speed

Fast pulses control power (PWM)



Timers

Delays for lights and alarms

Big idea

Not every problem needs a microcontroller. A simple chip is **cheaper**, uses **less power** and **never crashes**. Good engineers choose the simplest tool that works.

9
Industry & Innovation

Next week

YI-02: a heart rate monitor with a sensor and OLED screen.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. How fast will it blink?
2. Test it in the Falstad simulator
3. Breadboard steps
4. Make it your own
5. Bug hunter's guide

MAKE

Do the maths

How fast will it blink?

$$f = 1.44 / ((R1 + 2 \times R2) \times C)$$

f in hertz (Hz) = flashes per second · R in ohms · C in farads

Try it
Predict first, then test and time it with a stopwatch!

Worked example

R1 = 1,000 Ω, R2 = 10,000 Ω, C = 100 µF = 0.0001 F

f = 1.44 / (21,000 × 0.0001) = 1.44 / 2.1 = **0.69 Hz**

About one flash every **1.5 seconds**.

R1	R2	C	Frequency	You see / hear
1 k	10 k	100 µF	**0.7 Hz**	Slow flash
1 k	10 k	10 µF	**6.9 Hz**	Fast beeping
1 k	5 k	10 µF	**13 Hz**	Rapid buzz

MAKE

Simulate first

Test it in the Falstad simulator

1. Open falstad.com/circuit and choose **Circuits → 555 Timer Chip → Square Wave Generator**.
2. Change R2 and C to match our circuit.
3. Watch the capacitor voltage rise and fall on the scope.
4. Turn the speed up and down. Does your formula agree?



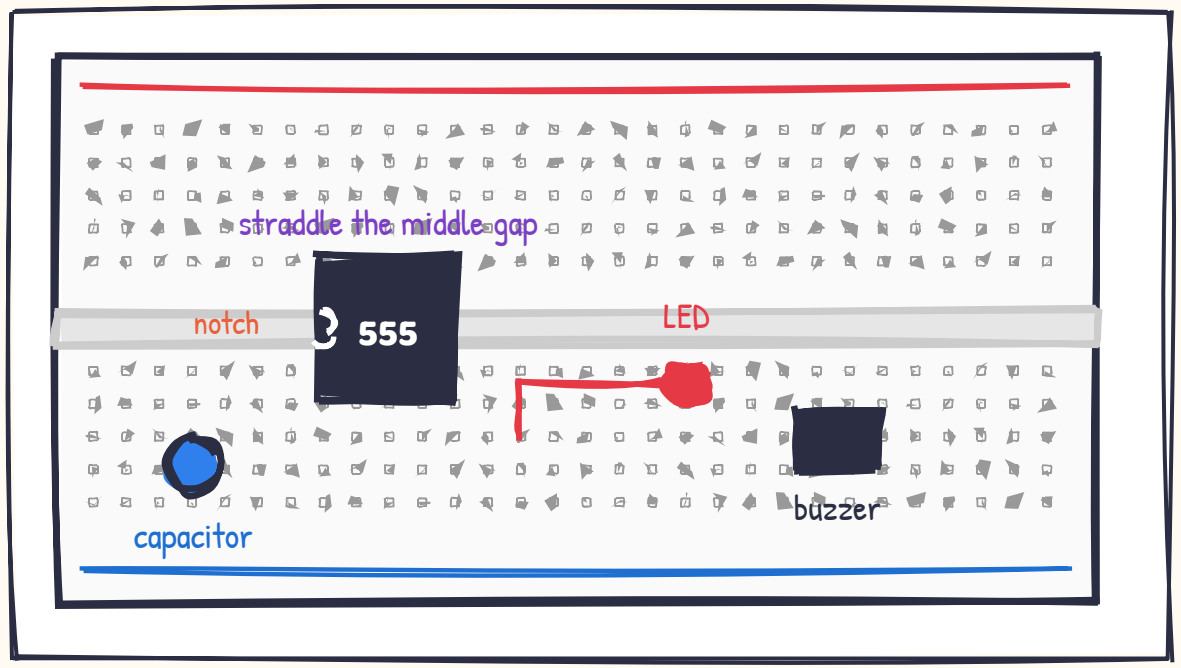
Why simulate?
Engineers test designs on a computer first. It is free, fast and nothing can burn!

MAKE

Build it

Breadboard steps

1. Place the 555 across the middle gap, notch on the left.
2. Pins 8 and 4 to the + rail; pin 1 to the - rail.
3. R1 from + to pin 7; R2 (pot) from pin 7 to pin 6.
4. Wire pin 6 to pin 2; capacitor from pin 2 to - (long leg to pin 2).
5. 10nF from pin 5 to -.
6. Pin 3 to LED (with 220 Ω) and buzzer, then to -.
7. Teacher check, then connect the 9V battery.



Watch out
Electrolytic capacitors have a + and – side. The stripe marks the – leg.

MAKE

Discovery challenges

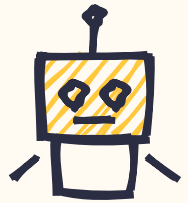
Make it your own

Speed knob

Turn R2. What happens to the flash rate? Explain why using the formula.

Swap the bucket

Change 100 µF to 10 µF. Predict the new frequency, then test.



Police lights

Add a second LED from + (with 220 Ω) to pin 3. The two LEDs now flash in turn!

Your own alarm

Design a rhythm for a bike alarm or a fire drill.

Why it works

When pin 3 is LOW, current flows from + through LED 2 into pin 3.

Test and debug

Bug hunter's guide

Nothing happens	Chip upside down? Check the notch. Is pin 4 connected to +?
LED stays on	Capacitor in the wrong way, or pin 2 and pin 6 not joined.
Very dim LED	Missing 220 Ω, or battery nearly flat: measure it.
Chip gets hot	Disconnect now! Usually a short from pin 3 or wrong power pins.

Engineer's habit
Unplug power before changing any wire.

Show what you know

Recap and exit ticket

Exit ticket

- [ ] Which pin is pin 1, and how do you find it?
- [ ] What two voltage levels make the 555 flip?
- [ ] What happens to the flash rate if C gets bigger?
- [ ] Calculate f for R1 = 1k, R2 = 10k, C = 10 µF.



Extension challenge

Use a second 555 to make a true two-tone siren: one slow timer switches a fast tone on and off.

Well done!
You built a timing circuit with no code, just physics!